Giacinto Paolo (GP) Saggese, PhD

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SUMMARY

- Technology executive, AI/ML researcher, and systems architect with 20+ years building AI/ML systems, quant strategies, and high-performance software
- Expertise in Causal AI, Bayesian modeling, knowledge graphs, optimization, and distributed systems
- 3x founder, team leader, and builder of production-grade ML and quant platforms

CORE COMPETENCIES

Causal AI | Bayesian Modeling | ML Infrastructure | Quant Finance Time-Series Forecasting | Knowledge Graphs | Distributed Systems | Algo Trading | Optimization | Product Strategy | Technical Leadership

EXPERIENCE

2023-today: CTO & Co-founder, *Causify.ai*

- Built a causal modeling platform using Bayesian inference, knowledge graphs, and temporal ML
- Designed automated pipelines for model building, scoring, and deployment
- Led technical vision, architecture, engineering, and early customer adoption
- Supported $5M fundraising and product strategy to $1M ARR

2023-today: Adjunct Professor, *University of Maryland*

- Instructor for "DATA605: Big Data Systems" (300 students per year) and "MSML610: Advanced ML" (120 students per year)
- Mentor for UMD Blockchain Accelerator and NSF I-Corps

2023-today: NSF I-Corps Instructor

- Train research teams to evaluate markets and develop commercialization paths

2021-2022: Portfolio Manager, *Engineers Gate* (New York City, NY)

- Built ML signals for equities and futures (minute-level horizon)
- Led 2-person research team; deployed models into production trading

2019-2020: CTO, *Particle.One* (San Jose, CA)

- Built ML models generating predictive commodity signals
- Raised $2.6M from Silicon Valley VCs
- Managed 10-person distributed team; delivered first commercial signals

2015-2019: Head of Alt-Data Group, *Teza Technologies* (Berkeley, CA)

- Developed ML/alt-data signals for futures, equities, FX, ETFs
- Built distributed pipelines for large-scale feature engineering and model training
- Led team of 12 researchers in Berkeley and Moscow

2011-2015: Co-founder & CTO, *June Inc.* (Santa Clara, CA)

- Designed quant strategies: stat-arb, market-neutral, long-short

- Built research stack: Python/R workflows, intraday simulator, portfolio optimizer, custom ML
- Wrote performance-critical C++ simulation components
- Raised $4M (Series A/B); led 8-person team

2010-2011: Senior Member of Technical Staff, *Synopsys*

- Integrated ZeroSoft's HPC simulation engine into Synopsys tools

2007-2010: Co-founder & VP Engineering, *ZeroSoft Inc.* (Milpitas, CA)

- Created simulation tech enabling **3-10x** faster IC verification
- Built compiler and graph-algorithm pipeline in C++/Verilog
- Raised $2M; customers included NVIDIA and Cisco
- Led 11 engineers; IP produced 2 patents
- Acquired by Synopsys for **$24M**

2005-2007: Senior Software Architect, *NVIDIA*

- Implemented C++ verification tools for Fermi GPU
- Designed GPU blocks in Verilog and FPGA

2004-2005: Postdoc, UIUC

2000-2004: PhD, Electrical & Computer Engineering, UIUC

- 20+ publications in microarchitecture, cryptography, and fault tolerance

1995-2000: Master's in Electrical Engineering, University of Naples

- GPA **3.99** / **4.00**, ranked 1st of 500+

TECHNICAL SKILLS

- **Programming:** Python, C++, GPU CUDA, x86 assembly
- **ML/AI:** Causal inference, Bayesian modeling, time-series ML, knowledge graphs, supervised/unsupervised ML, ML Ops
- **Quant:** Stat-arb, market-neutral strategies, portfolio optimization, alpha modeling
- **Systems:** HPC, distributed systems, GPU/FPGA design, compilers, simulators
- **Tools:** PyTorch, NumPy/SciPy, Pandas, Ray, Spark, Kubernetes (as relevant)

SELECTED TECHNICAL HIGHLIGHTS

- Built full-stack **Causal AI platform** (graph engine, Bayesian models, temporal ML, explainability)
- Designed **HPC simulation technology** delivering **3-10x** IC-verification speedup -> acquired by Synopsys
- Developed **alpha signal pipelines** for large unstructured/alt-data; deployed predictive signals across global markets
- Created **intraday simulators**, portfolio optimizers, and low-latency research infrastructure
- Led ML/engineering teams (8-12 people) across startups and quant firms
- Delivered **ML trading signals** with stable out-of-sample performance
- Built **ML Ops workflows**: data pipelines, feature stores, model validation, observability, CI/CD

ACHIEVEMENTS

- 20+ publications in top venues
- Granted US Patents:
  - 8,156,457, "Concurrent simulation of hardware designs with behavioral characteristics"
  - 8,738,350, "Mixed concurrent and serial logic simulation of hardware designs"
- Pending US Patents
  - "Systems and Methods for Causal Failure Prediction in Time Series" (U.S. Patent Application No. 63/934,790)
  - "Computational Systems and Methods for Time Series Data Processing" (U.S. Patent Application No. 63/960,355)
- O1 Visa recipient (extraordinary ability in sciences)
- Ranked 1st in class (500+ students)
- Under-18 Italian national swimming finalist (twice)

SELECTED PUBLICATIONS & PATENTS

- Research statement
- **Full list:** List of publication Google Scholar, DBLP ResearchGate
- **Patents**: US 8,156,457, US 8,738,350

OPEN SOURCE & CODE

GitHub Profile

- GitHub Stats

Causify.AI Projects

- **Causify Platform (csfy):** GitHub - Main causal AI platform implementing production-grade systems combining Bayesian inference, knowledge graphs, and temporal ML
- **Causify Helpers:** GitHub - Core utilities library for data processing, model utilities, and common operations
- **Causify Tutorials:** GitHub - Educational resources with Jupyter notebooks, example datasets, and step-by-step guides

LANGUAGES

- **English:** Native or bilingual proficiency
- **Italian:** Mother tongue
- **Citizenship:** USA / Italy